

ORIGINAL BASIC SCIENCE ARTICLE

Molecular aspects of collagenolysis associated with stress urinary incontinence in women with urethral hypermobility vs intrinsic sphincter deficiency

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Abstract

Aims: We aimed to assess the alterations in specific molecular mechanisms associated with collagenolysis and their correlation in cases of stress urinary incontinence (SUI) by urethral hypermobility (UH) and by intrinsic sphincter deficiency (ISD).

Methods: We evaluated the messenger RNA (mRNA) expression of matrix metalloproteinases (MMP1, MMP2, and MMP9), specific tissue inhibitors of metalloproteinases (TIMPs), and associated proteins (TIMP1, TIMP2, and α -2-macroglobulin [A2McG]) in the suburethral tissue of women with SUI (half with SUI by UH and half with SUI by ISD) by reverse-transcriptase quantitative polymerase chain reaction and determined the correlations of the expression of these proteins on each pathological condition.

Results: In the suburethral tissue, we found significantly altered *MMP2* mRNA levels, being higher in women with ISD than in those with UH. Still, *MMP1* and *MMP9* exhibited a tendency to a higher expression in women with ISD. Concerning the TIMPs expression, no significant statistical differences were found between women with UH and ISD. When evaluating the MMP/TIMP ratios, we found significant statistical differences between *MMP9/TIMP1* and *MMP9/A2McG*. In fact, the correlations of the expression *MMP9* with *TIMP1* and *A2McG* were lost for women with UH. The ratios between *MMP1/TIMP1* and *MMP1/A2McG* were similar, although there was a tendency of higher ratio in ISD for *MMP2/TIMP2* and *MMP2/A2McG*.

Conclusion: Our results point toward higher collagenolysis in the suburethral tissue of women with SUI by ISD.

KEYWORDS

collagenolysis, intrinsic sphincter deficiency, matrix metalloproteinases, stress urinary incontinence, urethral hypermobility

Abbreviations: A2McG, α -2-macroglobulin; ALPP, abdominal leak-point pressure; ECM, extracellular matrix; ICS, International Continence Society; ISD, intrinsic sphincter deficiency; MMPs, matrix metalloproteinases; MUCP, maximal urethral closure pressure; POP, pelvic organ prolapse; qRT-PCR, quantitative reverse-transcriptase polymerase chain reaction; SUI, stress urinary incontinence; TIMP, tissue inhibitor of metalloproteinase; UH, urethral hypermobility.

1 | INTRODUCTION

Stress urinary incontinence (SUI) is defined by the International Continence Society (ICS) as a leakage resulting from increased abdominal pressure. From Petros and Ulmsten (Integral Theory)¹ and DeLancey (Hammock hypothesis),² this occurs due to a lack of support that can result from damage or biochemical alterations on the pelvic connective tissue.^{3,4}

In the past, SUI in women was attributed to two main etiologic factors: urethral hypermobility (UH) and intrinsic sphincter deficiency (ISD). UH was considered to be caused by a deficient support of the urethra, dependent on levator ani muscles, endopelvic fascia, and its connection to the tendinous arch, whereas ISD was seen as a deficiency in urethral sphincter, dependent on smooth and striated muscles, and venous plexus of urethral length. In more recent days, the trend is to ignore the distinction between these two diagnoses.⁵ In the most recent joint report of International Urogynecology Association and ICS on the terminology for female pelvic floor dysfunction,⁵ the diagnosis for ISD is ignored. Several authors and surgeons advocate that the classification in SUI by UH and by ISD is only of historical importance, as these conditions are treated by similar procedures.^{6,7} In fact, it has been suggested that ISD is often associated with a certain degree of UH, leading to an overlap between these two conditions.⁶

Numerous studies have shown that the collagen quantity in endopelvic fascia or skin is significantly lower in women with SUI, when compared with continent women, and that this difference should not be due to collagen synthesis.⁸⁻¹⁰ It has also been suggested that, in women with SUI and pelvic organ prolapse (POP), the expression levels of matrix metalloproteinases (MMPs), the main extracellular matrix (ECM) enzymes in collagen degradation, and their tissue specific inhibitors (TIMPs) should play an important role in the collagen degradation of vaginal/pelvic floor tissue.¹¹ In fact, previous studies have shown that, in the vaginal tissue of women with SUI or POP, the messenger RNA (mRNA) levels of some MMPs are higher and the ratios MMPs/TIMPs are altered.¹²⁻¹⁴ Most studies described an increased expression of specific MMPs,^{13,14} providing evidence that abnormal connective tissue dynamics might contribute to SUI etiology.

Our hypothesis is that MMPs and TIMPs expression might be altered in SUI and that there are important biological differences on what collagenolysis concerns, in vaginal suburethral mucosa, between the two conditions that originate SUI (UH and ISD), which might correlate with their former classification.

2 | MATERIALS AND METHODS

2.1 | Chemicals

EZNA Total RNA Kit was purchased from Omega Bio-Tek (Norcross, GA). NZY M-MuLV Reverse Transcriptase random hexamer mix and dNTPs NZYSet were purchased from NZYTech (Lisbon, Portugal). KAPA SYBR FAST qPCR Master Mix was purchased from GRiSP (Porto, Portugal). Primers were purchased from STAB VIDA (Oeiras, Portugal). Other chemicals were all purchased from Sigma-Aldrich (St Louis, MO), unless stated otherwise.

2.2 | Patient selection and ethical issues

In a universe of 120 patients with SUI submitted to surgery in the urogynecology consultation, 30 patients who meet the inclusion and exclusion criteria (as described below) were selected (15 with SUI by UH and 15 SUI by ISD). All patients filled a questionnaire that was further confirmed by physical examination and by urodynamic study with SUI criteria for both entities. The presence of SUI is multifactorial, so the sample is defined as postmenopausal women with SUI and with vaginal deliveries with no other risk factors, who have not been submitted to hormonal replacement therapy and none of the exclusion criteria: anesthetic or surgical procedures contraindications, previous surgery for the treatment of urinary incontinence, vaginal, psychiatric disease, toxic consumption, or other conditions that compromise the ability to give informed consent or that may affect the bladder compliance. Moreover, women with uncontrolled diabetes, taking medication that alters urinary function, or diagnosed with collagen diseases, were excluded as well as women in childbearing age, undergoing pelvic radiotherapy, or that have any direct family relationship with the investigators.

We classified the patients in two different groups ($n = 15$ patients per group) based on clinical findings and urodynamic values. The first group presented UH (defined by more than 30° using Q-tip test), with abdominal leak-point pressure (ALPP) above 90 cmH₂O with maximal urethral closure pressure (MUCP) above 20 cmH₂O and the second group did not have UH, with ALPP below 90 cmH₂O but in preference below 60 cmH₂O with MUCP below 20 cmH₂O.¹⁵

All patients were elected for surgery of SUI, with suburethral sling by transobturator route. They have all given written consent for the procedure and for the inclusion on this study, that was approved by the ethical

committee of our hospital and was conducted in accordance with the Helsinki Declaration and National Legislation.

Surgeries were scheduled and performed by the same surgeon. During the procedure, a vaginal suburethral mucosa sample of about 1 cm³ was collected and was immediately transported to the research laboratory for processing.

2.3 | Quantitative reverse-transcriptase polymerase chain reaction

The extraction of total RNA of the samples was performed using the EZNA Total RNA Kit and its concentrations were determined by Qubit 2.0 Fluorometer (Invitrogen, Carlsbad, CA). RNA was reversely transcribed to resultant complementary DNA by routine methods, as previously described.¹⁶

The mRNA expression levels of three extracellular matrix metalloproteinases (*MMP1*, *MMP2*, and *MMP9*) and their endogenous inhibitors (*TIMP1*, *TIMP2*, and *A2McG*) were evaluated in each biopsy specimen using the quantitative reverse-transcriptase polymerase chain reaction (qRT-PCR) technique. qRT-PCR was performed in a StepOne (Applied Biosystems, Foster City, CA) system using specific exon-spanning primers to analyze *MMP1* (sense: CCAGGTATTGGAGGGGATGC; antisense: CGATGGGCTGGACAGGATTT), *MMP2* (sense: CACCTACACCAAGAACTTCC; antisense: AACACAGCCTTCTCCTCCTG), *MMP9* (sense: GCTTTTCTTCTTCTCTGGGCG; antisense: GGTCCTG GCAGAAATAGGCTT), *TIMP1* (sense: ATCCGGTTC GTCTACACCCC; antisense: TCCGTCCACAAGCAAT GAGTG), *TIMP2* (sense: CAAAGCGGTCAGTGAGA AGGA; antisense: CCTTCTTCTCCTCAACGTCCA), and *A2McG* (sense: ACAGAGCAGCATAAAGCCCA; antisense: TCTCAGTGGTCTCAGTGTGGA). Amplification conditions comprised an initial denaturation step of 3 minutes at 95°C, followed by 35 runs of a three-step cycle: a denaturation step of 3 seconds at 95°C; an annealing step of 30 seconds with a specific temperature for each set of primers; and an extension step of 10 minutes at 72°C. Primer's optimal annealing temperature of 56°C for *MMP2*, 58°C in the case of *MMP1* and *A2McG*, and 59°C for *MMP9*, *TIMP1*, and *TIMP2*, were required for exponential amplification phase of fragments. The quantification cycle (C_q) was determined through cycle threshold and fold variation of the expression levels was calculated following the Pfaffl mathematical model,¹⁷ using β -2-microglobulin as reference gene (sense: GCTTCAGTCGTCAGCA TGGC; antisense: GGATTTCAATGTGAGGCGGGT; optimal annealing temperature of 58°C).

2.4 | Statistical analysis

Statistical analysis was performed using the GraphPad 7 Software (GraphPad, San Diego, CA). The normality of data was evaluated using the Kolmogorov-Smirnov test and all data and variables were normally distributed (considering $P < 0.05$). Statistical differences between the experimental groups were assessed by the unpaired two-tail Student t test. No outliers were identified using the ROUT Method ($Q = 5.0\%$). All experimental data are shown as mean $\pm$ SEM ($n = 15$ for each condition) and $P < 0.05$ was considered significant. Data correlation was evaluated by assuming a two-tailed Gaussian distribution, with a confidence interval of 95%.

3 | RESULTS

3.1 | Patient selection

A total of 30 patients, of which 15 presented SUI by UH and 15 were diagnosed with SUI by ISD, were selected for this study and classified according to ALPP (above or below 90 cmH₂O; Table 1). No differences were found between

TABLE 1 Urodynamic characterization of patients from the urethral hypermobility (UH) and intrinsic sphincter deficiency (ISD) groups ($n = 15$ each)

Sample characterization	UH	ISD	P value
Age range, y	52-70	52-81	
Mean age, y	59.2	58.0	0.71
Menopausal age range, y	46-56	42-55	
Mean menopausal age, y	49.1	46.6	0.22
Mean parity	2	2	1
Mean body-mass index, kg/m ²	29.64	27.35	
Physical examination			
Leak with maneuvers (Yes/No)	13/2	15/0	
Urethral hypermobility (Yes/No)	12/3	0/15	
Urodynamic studies			
Capacity range, mL	298-526	343-600	
Mean capacity, mL	381.7	428.7	0.36
MUCP			
Maximum urethral closure pressure range, cmH ₂ O	90-152	19-65	
ALPP			
Abdominal leak-point pressure range, cmH ₂ O	92-174	60-89	
Results of King's Health Questionnaire			
Median score	50	72	

Abbreviations: ALPP, abdominal leak-point pressure; MUCP, maximal urethral closure pressure.

All P values greater than 0.05 were considered not statistically significant.

the two groups in what concerns to the age (59.20 ± 2.22 vs 58.08 ± 3.82 years, respectively), menopausal age (49.15 ± 1.27 vs 46.65 ± 3.26 years, respectively), body-mass index (29.64 ± 1.62 vs 27.35 ± 1.84 kg/m², respectively), capacity (381.7 ± 57.8 vs 428.7 ± 61.5 mL, respectively), and parity (2.00 ± 0.30 vs 2.02 ± 0.35 , respectively; Table 1).

3.2 | MMP2 expression is increased in patients with ISD

Concerning the expression of MMPs, our results showed an increase of more than 100% in *MMP2* mRNA expression ($P = 0.042$) in patients with ISD (0.36 ± 0.14 arbitrary units), when compared to the UH group (0.17 ± 0.08 arbitrary units). However, although there was a tendency for an increase in the *MMP1* mRNA expression on the suburethral tissue of women with ISD, there was no significant difference between patients with UH (0.11 ± 0.05 arbitrary units) and ISD (0.22 ± 0.08 arbitrary units). Moreover, there were no differences concerning *MMP9* mRNA expression when comparing patients with UH (0.0015 ± 0.0005 arbitrary units) with patients with ISD (0.0079 ± 0.0047 arbitrary units), though *MMP9* had also a higher expression in ISD (Figure 1).

3.3 | TIMPs expression is similar in patients with UH and ISD

The mRNA expression levels of *TIMP1* were not statistically different between ISD (0.17 ± 0.05 arbitrary units) and UH patients (0.11 ± 0.03 arbitrary units) (Figure 2). Concerning *TIMP2* and *A2McG*, although the differences did not reach statistical significance, their mRNA expression levels were more elevated in the

suburethral tissue of women with ISD (0.093 ± 0.046 and 0.100 ± 0.045 arbitrary units, respectively) as compared to that of women with UH (0.032 ± 0.006 and 0.047 ± 0.020 arbitrary units, respectively; Figure 2).

3.4 | Patients with ISD show higher proteolytic activity

The ratio of MMP to TIMP mRNA expression may be used as a marker of proteolytic activity, with a higher ratio showing a higher tendency for proteolytic activity in that specific tissue. Our results show a significant increase in *MMP9/TIMP1* ($P = 0.0045$; Figure 3A) and *MMP9/A2McG* ($P = 0.0072$; Figure 3B) ratios for patients with ISD (0.030 ± 0.011 and 0.087 ± 0.018 arbitrary units, respectively), when compared to patients with HU (0.013 ± 0.002 and 0.058 ± 0.016 arbitrary units, respectively). On the other hand, there was no statistically significant difference concerning *MMP1/TIMP1* ratio when comparing the patients with UH (0.69 ± 0.19 arbitrary units) and ISD (0.84 ± 0.22 arbitrary units) (Figure 3A). The ratio *MMP1/A2McG* was also similar in both groups (UH: 2.42 ± 0.54 ; ISD: 2.52 ± 0.64 arbitrary units). On the other hand, there was a tendency of higher ratio *MMP2/TIMP2* (9.12 ± 4.98 arbitrary units) and *MMP2/A2McG* (3.92 ± 0.71 arbitrary units) in patients with ISD, when compared to the UH group (4.27 ± 1.71 and 2.46 ± 0.21 arbitrary units, respectively).

3.5 | TIMP1 expression is highly correlated with MMPs expression

Our results show strong positive correlations between the MMPs expression, both in UH and ISD patients,

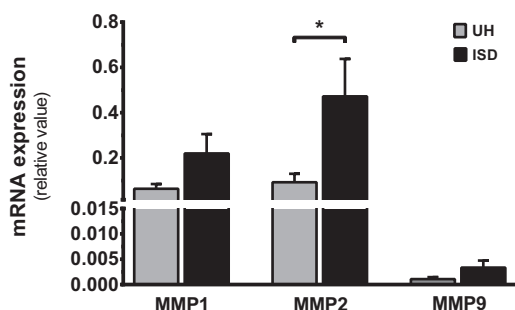


FIGURE 1 Matrix metalloproteinases (MMPs) messenger RNA (mRNA) expression in vaginal suburethral mucosa. The figure shows pooled data of independent experiments, indicating *MMP1*, *MMP2* and *MMP9* mRNA expression levels found in vaginal suburethral mucosa of patients with stress urinary incontinence (SUI) by urethral hypermobility (UH) or by intrinsic sphincter deficiency (ISD). Results are expressed as mean $\pm$ SEM ($n = 15$ for each condition). Significant results are indicated by * $P < 0.05$; ** $P < 0.005$; *** $P < 0.001$

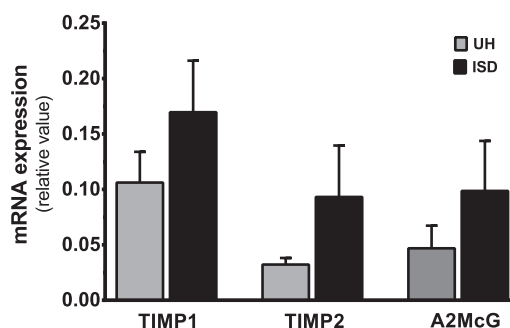


FIGURE 2 Matrix metalloproteinase inhibitors (TIMPs) messenger RNA (mRNA) expression in vaginal suburethral mucosa. The figure shows pooled data of independent experiments, indicating *TIMP1*, *TIMP2*, and α -2-macroglobulin (*A2McG*) mRNA expression found in vaginal suburethral mucosa of patients with stress urinary incontinence (SUI) by urethral hypermobility (UH) or by intrinsic sphincter deficiency (ISD). Results are expressed as mean $\pm$ SEM ($n = 15$ for each condition). Significant results are indicated by * $P < 0.05$; ** $P < 0.005$; *** $P < 0.001$

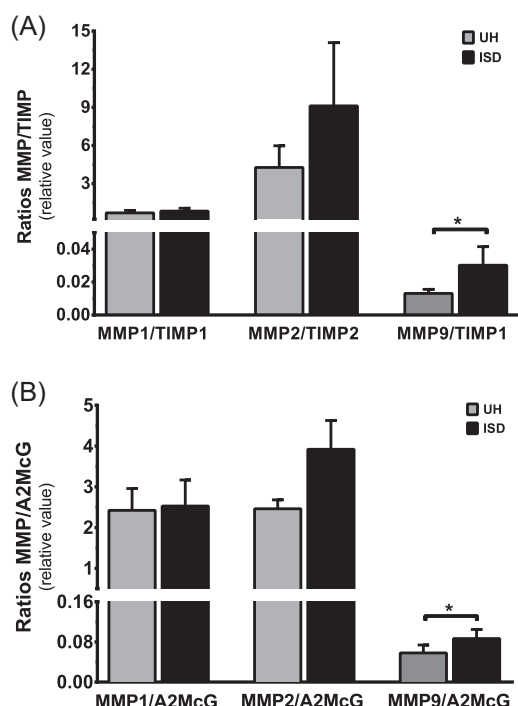


FIGURE 3 Ratios of matrix metalloproteinases (MMPs) and its inhibitors' (TIMPs) messenger RNA (mRNA) expression in vaginal suburethral mucosa. The figure shows pooled data of independent experiments, indicating the ratios between MMPs and TIMPs (a) and between MMPs and α -2-macroglobulin (A2McG) (b) mRNA expression found in vaginal suburethral mucosa of patients with stress urinary incontinence (SUI) by urethral hypermobility (UH) or by intrinsic sphincter deficiency (ISD). Results are expressed as mean $\pm$ SEM ($n = 15$ for each condition). Significant results are indicated by * $P < 0.05$; ** $P < 0.005$; *** $P < 0.001$

particularly concerning *MMP1* and *MMP2*. In ISD patients, *TIMP1* expression was strongly correlated with *MMP1* and *MMP9* (Table 2), while the correlation *MMP9-TIMP1* was lost for women with UH (Table 2). Similarly,

the correlation of *MMP9* with *A2McG* was lost for patients with UH, as compared with those with ISD. No significant alteration was observed in the correlation between *MMP2* and *TIMP2* in the samples from the individuals from both conditions.

4 | DISCUSSION

MMPs together with their TIMPs are involved in ECM remodeling, which is a complex combination of structural proteins, including collagen and elastin.^{18,19} MMPs play an important role in the degradation of vaginal/pelvic floor conjunctive tissue.¹³ The tensile strength of this tissue is maintained by continuous remodeling of the ECM, with the synthesis and degradation of ECM components being equally important. Collagen types I, III, and V are the main structural components of vaginal/pelvic floor tissue. *MMP1* cleaves these interstitial collagens, while the cleaved collagen fragments are susceptible to rapid degradation by *MMP2* and *MMP9*. These metalloproteinases (*MMP2* and *MMP9*) also degrade elastin,^{12,18} which is closely connected with collagen fibers, contributing to the tensile strength of the vaginal/pelvic floor tissue. In contrast to collagen, elastin is rarely synthesized in the ECM tissue of adults, being that its repair frequently results in dysfunctional products.¹¹ Similar to what happens with collagen, there is also evidence of increased elastin degradation in SUI cases.¹¹

TIMPs bind stoichiometrically to MMPs to inhibit their activity. The balance between MMPs and TIMPs defines collagenolysis and the homeodynamics of ECM.¹³ The activity of *MMP2* is dependent of *TIMP2*. *MMP9* and *MMP1* are inhibited by *TIMP1*. *A2McG* has been

TABLE 2 The Pearson correlation coefficients between the mRNA expression levels of matrix metalloproteinases (MMPs) and of specific tissue inhibitors of metalloproteinases (TIMPs), in the suburethral tissue of women with stress urinary incontinence (SUI) by intrinsic sphincter deficiency (ISD) or by urethral hypermobility (UH)

		MMP2		MMP9		TIMP1		TIMP2		A2McG	
		R	P	R	P	R	P	R	P	R	P
ISD	MMP1	0.993	***	0.912	**	0.987	***	0.727	**	0.978	***
	MMP2	–	–	0.877	*	0.967	***	0.746	**	0.990	***
	MMP9	–	–	–	–	0.932	***	0.515	*	0.867	***
UH	MMP1	0.960	***	0.729	**	0.847	***	0.706	**	0.862	***
	MMP2	–	–	0.588	*	0.914	***	0.557	*	0.932	***
	MMP9	–	–	–	–	0.481	NS	0.984	***	0.430	NS

Abbreviations: A2McG, α -2-macroglobulin; NS, nonsignificant.

Note: The association between MMPs and TIMPs expression levels was evaluated by computing the Pearson correlation coefficients (*R*) assuming a two-tailed Gaussian distribution and a confidence interval of 95%.

* $P < 0.05$; ** $P < 0.005$; *** $P < 0.001$. All *P* values less than 0.05 were considered statistically significant.

described as an extracellular proteinase inhibitor with the unique ability to inhibit almost all known proteases, regardless of their mechanistic classes (among which are comprehended the MMPs).²⁰ It is therefore an inhibitor of MMP1, MMP2, and MMP9. Recent data suggests that A2McG is involved in ECM remodeling during pregnancy and that its decreased expression and/or activity in the vaginal wall tissues may contribute to the development of urinary incontinence.^{13,21}

During this study, we found that the mRNA expression patterns of the major players of conjunctive tissue remodeling were distinct in the vaginal/pelvic floor tissue samples from patients with ISD and UH. The expression of all the MMPs evaluated (*MMP1*, *MMP2*, and *MMP9*) showed a tendency to be higher in the samples from the patients with ISD, with the expression of *MMP2* reaching the statistical significance, as compared to that observed in the samples from the patients with UH. Concurrently, the expression levels of the MMP inhibitors (*TIMP1*, *TIMP2*, and *A2McG*) also showed an increasing tendency in the ISD samples, although there was no statistical difference from the samples obtained from UH patients. In fact, it has already been reported that some conditions associated with altered collagenolysis in samples of vaginal tissue (e.g. POP) are characterized by an increased tissue expression of *MMP2*.²² Indeed, several studies have described that increased MMP levels are associated with reduced collagen content in various pelvic tissues or structures of women with POP.²³⁻²⁵ These findings, however, were not in agreement with others who found higher expression of *MMP1* or *MMP9* but not of *MMP2* in vaginal tissue from women with POP,^{12,13} though the methodology and the characteristics of the studied population used in the various studies were distinct.

Due to these discrepancies, we evaluated the MMPs/TIMPs ratios. These ratios are often used to more accurately evaluate the status of the pelvic ECM turnover.¹³ When analyzing the MMPs/TIMPs ratios, we found *MMP1/TIMP1* values were almost identical in the samples of both groups, suggesting no alterations in the *MMP1*-dependent remodeling of the ECM. In contrast, *MMP2/TIMP2* was higher in the samples of the ISD group which suggests that collagenolysis and/or elastinolysis may be increased in this group. The *MMP9/TIMP1* ratio was also higher in the ISD group, which further suggests an alteration of ECM homeodynamics dependent from this MMP and its inhibitor in the samples from the ISD group. Moreover, all ratios between the various MMPs and A2McG, were higher in the suburethral tissue of women with ISD, being that only the ratio *MMP9/A2McG* was statistically different between the samples of both groups. As *MMP2* and *MMP9* are gelatinases that degrade the products from collagen

cleavage and elastin, producing a weaker and disorganized ECM, the differential expression of these proteases or the unbalanced expression respectively to their specific inhibitors observed in the samples from the patients with ISD (as compared with the patients with UH) suggest a distinct etiology of these two pathological conditions.

5 | CONCLUSIONS

In this study, we investigated specific molecular mechanisms associated with collagenolysis in samples of suburethral tissue of women with SUI by UH and by ISD. Our results suggest that the etiology of SUI by UH or by ISD is not as simple as it has been described. A higher collagenolysis is demonstrated in the samples from women with ISD, which suggests that we cannot attribute the SUI to a mere sphincter dysfunction on these cases, and that the degradation of the ECM collagen may play an important role.

We need further studies to draw definite conclusions about the involvement of collagenolysis on SUI, particularly by ISD. This may imply a new way of approaching ISD. To find a way of counteracting the action of MMPs or elevate the expression of their specific tissue inhibitors may be an important field to investigate, to find alternatives for future treatment of women with these conditions.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

AUTHOR CONTRIBUTIONS

Project development: BC and PFO; data collection: BC, MJM, MGA, and PFO; data analysis: BC, MJM, MS, and PFO; manuscript writing: BC, MJM, MGA, MS, and PFO.

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